Same Company, Two Prices: How Closely Do India's Two Stock Exchanges Agree on the Close?

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Data and code: github.com/udit20112007/WORLD-CLOSE

Abstract

Thousands of Indian companies are listed on both the National Stock Exchange (NSE) and BSE, so each of them ends every trading day with two official closing prices. Using a year of daily closing data that I collected myself (2025-10-20 to 2026-10-26, 250 trading days), I match 2,524 companies across the two exchanges and study 501,497 pairs of closing prices. The two closes are usually close but rarely identical: the median gap is 11.0 basis points, and 13.7% of stock-days differ by more than half a percent. Liquidity explains most of the variation across companies. A tenfold increase in a stock's trading value is associated with a median gap about 54% smaller, and trading value alone accounts for 84% of the cross-sectional variation in typical gaps. The gaps do not last: only about 14% of a gap carries over to the next day on average, and for the most actively traded stocks almost none does. Gaps widen modestly on days when the market moves a lot. I also document a puzzle: for thinly traded stocks, BSE tends to close slightly above NSE. Overall, the evidence fits a market where the law of one price holds well for liquid stocks, and where the gaps that remain look like short-lived noise rather than lasting mispricing.

1. Introduction

The law of one price is one of the simplest ideas in finance: the same asset should not sell for two different prices at the same time. If it did, a trader could buy where it is cheap, sell where it is dear, and pocket the difference. In practice, the idea is tested whenever the same security trades in more than one place, and a long line of research shows that the law can bend. Twin shares such as Royal Dutch and Shell traded for years at prices inconsistent with their fixed claim on the same profits (Froot and Dabora, 1999), and parent companies have at times been valued below the stakes they held in their own subsidiaries (Lamont and Thaler, 2003). The usual explanation is that arbitrage is costly and risky, so small or awkward deviations can survive (Shleifer and Vishny, 1997).

India offers an unusually clean setting for asking the question on a large scale. Most large and mid-sized Indian companies are listed on both NSE and BSE. The shares are the same security, identified by the same ISIN, settled through interoperable clearing, and owned by the same kind of investors. Unlike twin shares or cross-listings in different countries, there is no exchange-rate risk, no difference in time zone, and no conversion between share classes. If the two closing prices differ, the difference has to come from how and where the stock happened to trade that afternoon.

This paper asks three simple questions. How large are the gaps between NSE and BSE closing prices? Which stocks have larger gaps? And do gaps close, or do they persist from one day to the next? To answer them I built an automated data pipeline that downloads each exchange's official end-of-day file every trading day, and I use one year of that data.

The main findings are easy to state. First, gaps are small but common: the median absolute gap is 11.0 basis points (0.11%), and the two closes are exactly equal on only 3.3% of stock-days. Second, liquidity is the dominant explanation. Across companies, the typical gap falls steadily as trading activity rises, from about 50 basis points for the least-traded fifth of companies to about 4.5 for the most-traded fifth. Third, gaps are short-lived. On average only 14% of today's gap shows up again tomorrow, and after a gap larger than half a percent the next day's gap is smaller 78% of the time. Fourth, there is a small but systematic tendency for BSE to close above NSE in thinly traded stocks, which I cannot fully explain with this data.

Section 2 describes the institutional setting. Section 3 explains the data and how the sample is built. Section 4 presents the results, Section 5 checks their robustness, and Section 6 asks whether the gaps are large enough to trade on. Section 7 discusses limitations, and Section 8 concludes.

2. Institutional background

NSE and BSE are India's two main stock exchanges, both based in Mumbai. For companies listed on both, NSE handles the great majority of trading. In my sample, BSE accounts for 7.0% of the combined value traded in dual-listed stocks, and the median company does only 5.5% of its trading on BSE. This matters for interpretation: BSE's closing price is usually based on far fewer trades than NSE's.

Neither exchange uses the very last trade of the day as its official closing price. As I understand the exchanges' published methodologies, both compute the official close as a weighted average of trades in the final half hour of continuous trading. The close is therefore an average over a window, and two exchanges can arrive at different averages simply because different trades happened on each during that window. A stock that trades only a handful of times on BSE in that half hour will have a BSE close that reflects just those few trades, including whether they happened to execute at the bid or the ask. This mechanical source of difference is important when interpreting the results: a gap between the two closes is not, by itself, evidence of a price at which anyone could have traded.

Moving shares between the exchanges is straightforward. Since 2019, the clearing corporations of the exchanges have been interoperable, so a position bought on one exchange can be sold on the other. This makes the setting close to a textbook case: the same security, the same settlement, two prices.

3. Data and sample

The data come from the official daily "bhavcopy" files that both exchanges publish after each session. Each file lists every stock that traded that day, with its open, high, low and official closing price, the previous close, the number of shares traded and the value traded. I wrote a pipeline that downloads both files every trading day, standardises their columns, and stores them; a year of history was filled in from each exchange's archive. The same pipeline powers a public website that shows every stock's close on 33 exchanges worldwide, but this paper uses only the two Indian exchanges.

Companies are matched across the two exchanges by ISIN, the international identifier that is the same for a given security wherever it trades. Because each exchange's file only includes stocks that actually traded, a matched pair means the stock traded on both exchanges that day. I then apply two filters, shown in Table 1. I keep only NSE's regular equity series (EQ). This removes stocks in restricted segments, such as those that may only be traded for delivery (BE) or that belong to companies that have not met listing requirements (BZ). I also

drop stocks priced below Rs 20, where the minimum price step of one paisa is large relative to the price and a one-tick difference becomes a big percentage gap.

Table 1. Sample construction

Step	Companies	Stock-days
All companies matched by ISIN on both exchanges	2,848	595,598
NSE regular equity series (EQ) only	2,751	550,726
Share price of at least Rs 20	2,524	501,497

The filters matter more than their size suggests. In an early version of the analysis without them, the gap on one day was strongly correlated with the gap on the next (a correlation of 0.86), which seemed to imply that gaps were extremely persistent. Looking at the companies behind that result showed that it was driven almost entirely by a few restricted penny stocks with huge, stable gaps, some of them priced well below one rupee. Once these were removed, the correlation fell to 0.15. I report this because it is a good example of how a few unusual securities can dominate a statistic computed over half a million observations.

4. Results

For each stock and day, I define the gap as the BSE close minus the NSE close, divided by the average of the two, and express it in basis points (one basis point is one hundredth of a percent). A positive gap means BSE closed higher. Most of the analysis uses the absolute gap, which measures disagreement regardless of direction.

4.1 How large are the gaps?

Table 2 summarises the distribution. The typical gap is small, 11.0 basis points, but the distribution has a long tail: the mean is 29.0 basis points, almost three times the median, and one stock-day in twenty has a gap above 111 basis points. The two closes are identical on only 3.3% of stock-days, which is what one would expect if each close is an average of different trades. Across the whole sample there is no tendency for either exchange to close higher: BSE is above NSE on 49.2% of stock-days and the median signed gap is essentially zero. Figure 1 shows the shape of the distribution, which is symmetric, sharply peaked at zero and heavy-tailed.

Table 2. The size of NSE–BSE closing-price gaps (clean sample)

Statistic	Value
Median absolute gap	11.0 bps
Mean absolute gap	29.0 bps
75th / 90th percentile	27.9 / 65.4 bps
95th / 99th percentile	111.2 / 293.8 bps
Share of stock-days with identical closes	3.3%
Share with gap above 10 / 25 bps	52.8% / 27.6%
Share with gap above 50 / 100 / 200 bps	13.7% / 5.8% / 2.0%
Share of stock-days where BSE closed higher	49.2%

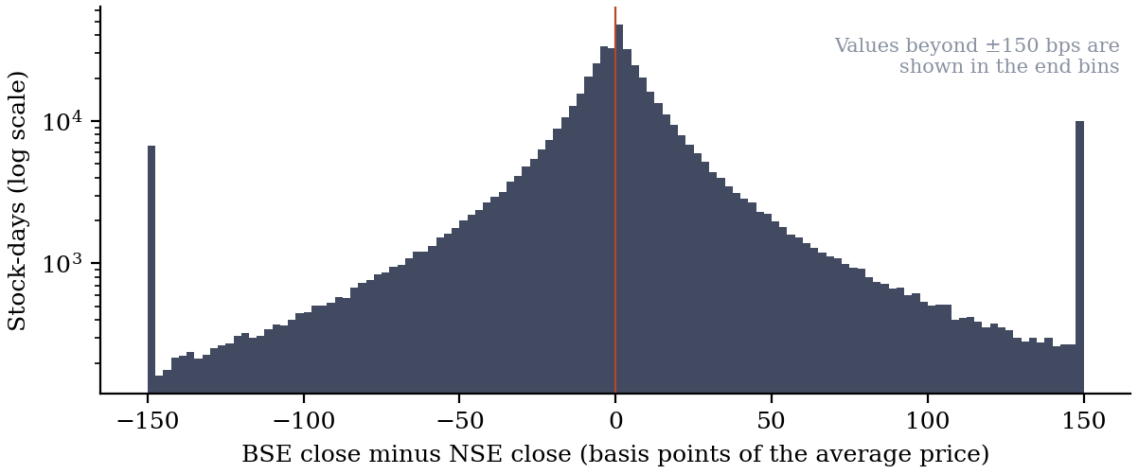


Figure 1. Distribution of the signed gap between BSE and NSE closing prices, in basis points of the average of the two prices. The vertical axis is logarithmic, so the tails are visible. Values beyond ±150 basis points are collected in the end bins.

4.2 Which stocks have larger gaps?

To see which stocks have larger gaps, I summarise each company by its median absolute gap over the year and relate it to its characteristics. Table 3 groups companies into fifths by how much they trade, and the pattern is striking. The typical gap falls at every step, from 50.2 basis points in the least-traded fifth to 4.5 in the most-traded fifth, a difference of about 11 times. Figure 2 shows the same relationship company by company. On logarithmic scales it is close to a straight line, which means the gap shrinks by a roughly constant percentage for each doubling of trading activity.

Table 3. Typical gaps by liquidity group (companies with at least 60 matched days)

Liquidity group	Companies	Median daily value traded on NSE	Median absolute gap
Least-traded fifth	443	Rs 0.10 crore	50.2 bps
2nd fifth	442	Rs 0.84 crore	20.6 bps
3rd fifth	442	Rs 3.86 crore	11.5 bps
4th fifth	442	Rs 13.94 crore	7.5 bps
Most-traded fifth	443	Rs 84.11 crore	4.5 bps

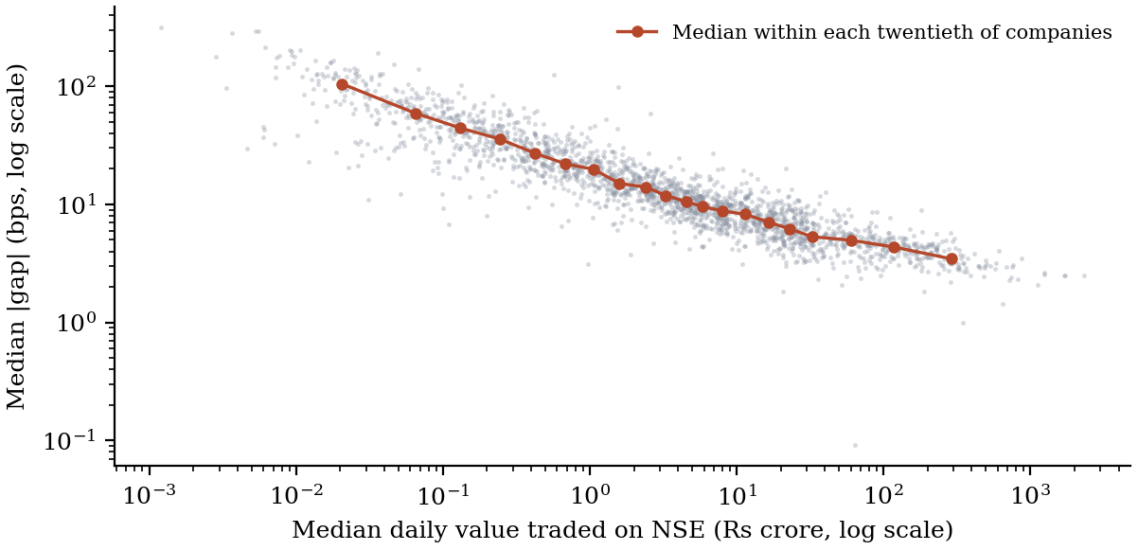


Figure 2. Each grey point is one company: its median daily value traded on NSE against its median absolute gap over the year. The line joins the medians within twenty equal-sized groups of companies. Both axes are logarithmic.

Table 4 reports a regression of each company's (log) median gap on four characteristics: how much it trades, how much of that trading happens on BSE, how volatile it is, and how large one price step is relative to its price. Standard errors are robust to heteroskedasticity. The model explains 88% of the variation in typical gaps, and trading value on its own explains 84%. The coefficient on log trading value is -0.34: a company that trades ten times as much has, other things equal, a typical gap about 54% smaller. Volatility pushes the other way: each extra percentage point of daily volatility is associated with a gap about 27% larger, which makes sense, since prices move more within the closing window of a volatile stock. A larger BSE share of trading goes with smaller gaps (about 7% smaller for ten extra percentage points), consistent with the idea that BSE's close is noisier when it rests on fewer trades. The relative tick size has a small negative coefficient. Its sign is the opposite of what I expected, and since prices below Rs 20 have already been removed, I do not read much into it.

Table 4. What explains a company's typical gap? Dependent variable: $\log_{10}$ of the company's median absolute gap (bps). OLS with heteroskedasticity-robust standard errors.

Variable	Coefficient	Std. error	t-statistic
$\log_{10}$ median daily value traded on NSE	-0.3413	0.0037	-91.1
BSE share of trading (%)	-0.0031	0.0010	-3.3
Daily return volatility (%)	0.1030	0.0047	22.0
Relative tick size (100 / price)	-0.0156	0.0040	-3.9
Constant	3.4423	0.0350	98.2
Companies / R^2	2,212		0.882

4.3 Do gaps close?

If gaps were genuine mispricing that took time to correct, today's gap would predict tomorrow's. I test this by regressing each stock's gap on the next trading day on its gap today. The slope is 0.139 (standard error 0.016, clustered by company, 497,948 observations). In other words, only about 14% of a gap carries over to the next day on average, which corresponds to a half-life of well under a day. Persistence is highest for the least-traded stocks, with a slope of 0.137, and falls to 0.012 for the most-traded fifth, where

tomorrow’s gap is essentially unrelated to today’s.

Large gaps behave the same way. On the 64,160 stock-days with a gap above 50 basis points, the next day’s gap was smaller 78% of the time, it shrank by a median of 58%, and it changed direction 48% of the time. That last number is telling: a gap that flips sign about half the time looks much more like random noise around zero than like a mispricing waiting to be corrected. Figure 3 shows this directly. For all three liquidity groups, the average gap tomorrow is nearly flat whatever today’s gap was.

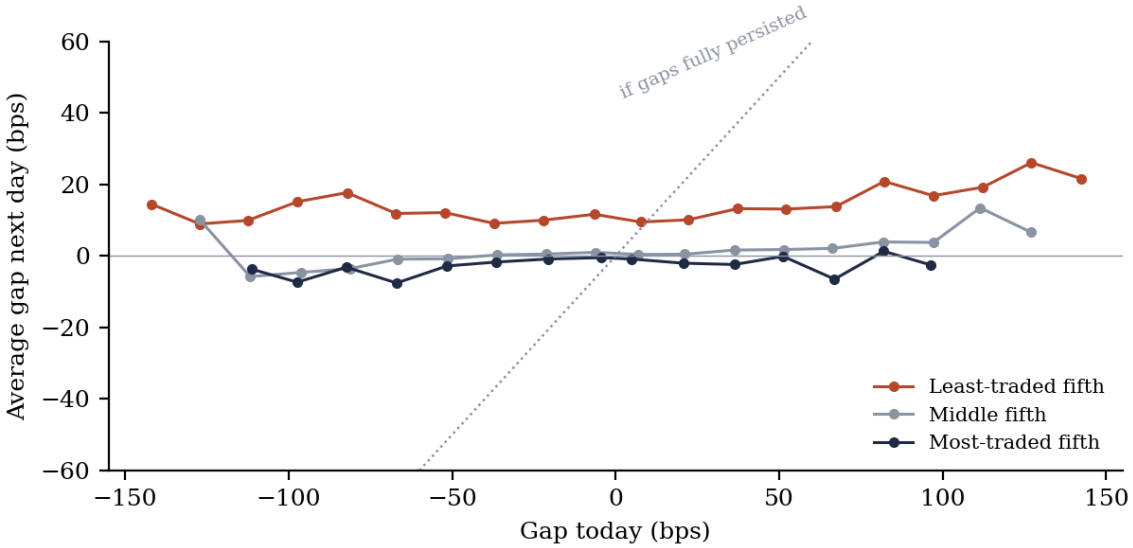


Figure 3. Average gap on the next trading day against today’s gap, for three liquidity groups. The dotted line shows what complete persistence would look like. All three groups are close to flat; the least-traded group sits slightly above zero throughout, reflecting the level effect discussed in Section 4.4.

4.4 A puzzle: BSE closes higher for thinly traded stocks

Figure 3 reveals something the absolute gap hides. For the least-traded stocks, the average gap is positive whatever happened the day before, which means BSE tends to close slightly above NSE. Table 5 shows this is systematic. In the least-traded fifth, the mean signed gap is +16.3 basis points and BSE closes higher on 54.2% of stock-days; 75% of these companies have a positive median gap over the year. The effect fades steadily with liquidity and reverses slightly for the most-traded stocks, where BSE closes higher on only 44.0% of days.

Table 5. The direction of the gap (BSE minus NSE) by liquidity group

Liquidity group	Mean signed gap	Median signed gap	BSE closed higher
Least-traded fifth	+16.3 bps	+6.5 bps	54.2%
2nd fifth	+1.9 bps	+1.0 bps	50.5%
3rd fifth	+0.7 bps	+0.4 bps	50.1%
4th fifth	+0.2 bps	+0.0 bps	48.3%
Most-traded fifth	-0.8 bps	-0.3 bps	44.0%

I do not have a definitive explanation, and the data cannot separate the possibilities. Three candidates seem worth investigating. The first is the composition of trading: if the few trades in illiquid stocks on BSE late in the day are more often buyer-initiated, they would execute nearer the ask and lift BSE’s average. The second is the closing-price rule itself: when very few trades occur on BSE in the final half hour, the way the exchange computes or substitutes the close could differ from NSE’s in a way that has a direction. The third is selection: a stock only enters the sample on days it traded on BSE, and the days when it trades there may not

be typical days. Telling these apart would need intraday trade data, which is beyond this paper.

4.5 Do gaps widen when the market is under stress?

Limits-to-arbitrage arguments suggest that gaps should widen when markets are volatile, because prices move faster within the closing window and arbitrage capital is stretched. To check, I compute the median absolute gap across all stocks for each day and compare it with the size of that day's market move, measured as the absolute value of the median stock's return. The two are positively related (correlation 0.35). A regression with standard errors that allow for correlation across nearby days gives a slope of 0.8 basis points of gap for each percentage point of market movement (t = 5.1). The effect is real but modest. On the calmer half of days the typical gap was 10.4 basis points, and on the most turbulent tenth it rose to 12.6. Figure 4 shows the daily series.

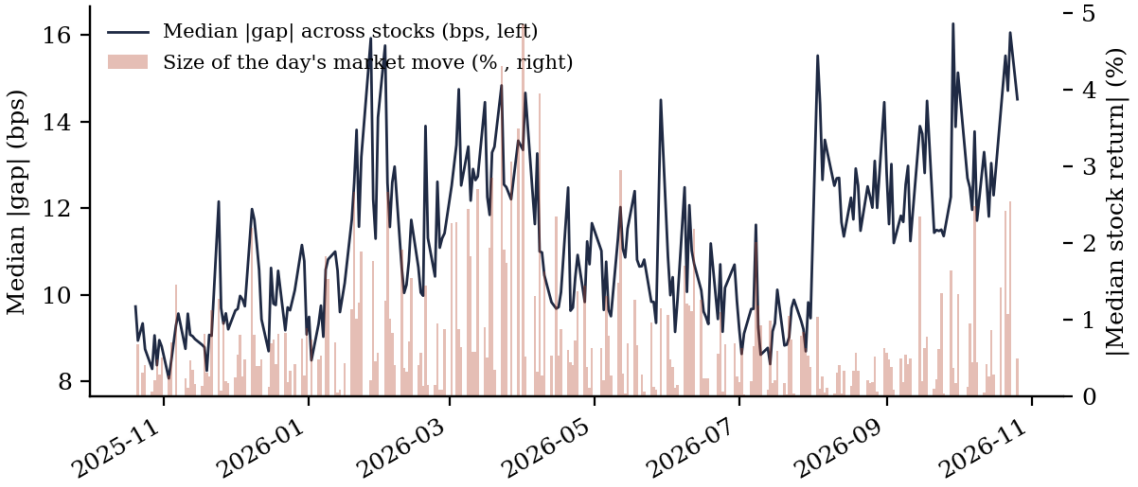


Figure 4. The line is the median absolute gap across all stocks on each day (left axis). The bars show the size of that day's market move, measured as the absolute return of the median stock (right axis).

5. Robustness

Table 6 repeats the two central results on different cuts of the data. Splitting the year in half gives almost identical answers in each half: the liquidity relationship has a slope of about −0.33 to −0.36 and explains around 80% of the variation, and the next-day persistence is the same in both halves. Using each company's mean gap rather than its median gives the same picture, as does capping extreme gaps before measuring persistence, which if anything lowers it further. The results do not depend on a particular period or on a few extreme observations.

Table 6. Robustness checks. The liquidity slope is from a regression of $\log_{10}$ median gap on $\log_{10}$ trading value alone, so every row is comparable.

Check	Liquidity slope	R^2	Persistence (next-day slope)
Full year (main sample)	-0.347	0.841	0.139
First half of the year (2025-10-20 to 2026-04-29)	-0.360	0.843	0.139
Second half of the year (2026-04-30 to 2026-10-26)	-0.325	0.794	0.139
Mean instead of median gap per company	-0.315	0.785	—
Gaps capped at ±200 bps	—	—	0.070

6. Is there money on the table?

A natural question is whether these gaps could be traded profitably. The evidence suggests not, for three reasons. First, the gaps are smallest exactly where trading is easiest. Among the most-traded fifth of companies, a gap above 50 basis points occurs on only 1.9% of stock-days, and above 100 basis points on 0.5%. The larger gaps are concentrated in stocks where trading a meaningful amount would itself move the price. Second, the closing price is an average over a half-hour window, not a price anyone could trade at, so observing a gap after the fact does not mean a trader could have captured it. Third, a round trip involves brokerage, exchange fees and taxes on both legs, and once those are counted the typical gap of around 11 basis points looks small. Combined with the finding that gaps largely disappear by the next day, the picture is of a market that arbitrageurs keep in line for the stocks where it matters, with the remaining differences reflecting noise in how each exchange's close is formed.

7. Limitations

The study has several limitations. It uses one year of data, so it cannot say whether the patterns are stable across market regimes or how they changed with past reforms. It uses only official closing prices; with intraday trades and quotes, one could measure gaps at the same instant, separate bid-ask effects from genuine disagreement, and resolve the BSE puzzle in Section 4.4. Prices are not adjusted for corporate actions, which does not affect same-day gaps between exchanges but does affect the volatility measure; I reduce this by excluding daily moves larger than 25%. The market-stress measure is a simple proxy based on the median stock's return rather than an index. Finally, the description of each exchange's closing-price rule is based on my reading of their published methodologies, and readers interested in the mechanism should consult the exchanges' current rules directly.

8. Conclusion

For more than two thousand Indian companies, two exchanges publish a closing price every day. Those two prices usually agree to within a tenth of a percent, rarely agree exactly, and disagree most for the stocks that trade least. The disagreements do not last: they largely vanish by the next day and switch direction about as often as not, which is what one would expect if they mostly reflect the particular trades that happened to land in each exchange's closing window. In that sense, the law of one price holds well where it is easiest to enforce. The small, persistent tendency for BSE to close higher in thinly traded stocks is the one pattern that does not fit neatly, and it would be a natural starting point for work with intraday data.

Data and code

The data were collected with an automated pipeline that downloads both exchanges' official daily files and stores them in a public repository, github.com/udit20112007/WORLD-CLOSE. The same data feed a public website, udit20112007.github.io/WORLD-CLOSE, which shows daily closing prices for 33 exchanges. The code that produces every number, table and figure in this paper is in the research/nse_bse_gaps folder of the repository, with instructions for rerunning it.

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Any errors are my own.

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